

Challenge (Habanero)

- Q1.** A recipe for making concrete requires $2 \cdot x$ pounds of cement and $3 \cdot x$ pounds of sand. If $x = 4$, how many pounds of sand are needed?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 16
- Q2.** A chef mixes $\frac{1}{4}$ cup of sugar with $\frac{3}{4}$ cup of honey to make a sauce. What is the ratio of sugar to honey?
(A) 1:2
(B) 1:3
(C) 1:4
(D) 2:3
(E) 3:4
- Q3.** A builder needs to make 18 gallons of paint that is 25% blue and 75% white. How many gallons of blue paint are needed?
(A) 3.6
(B) 4.5
(C) 6.0
(D) 9.0
(E) 12.0
- Q4.** A baker has two types of flour: type A costs 1.50 per pound and type B costs 2.25 per pound. If the baker wants to make a 50-pound mixture that is 30% type A and 70% type B, how much will the flour cost?
(A) 97.50
(A) 100.00
(A) 112.50
(A) 120.00
- (A) 130.00
- Q5.** A carpenter is making a wooden beam that is 12 feet long. The beam is made of two types of wood: type X and type Y. If 30% of the beam is type X and 70% is type Y, how many feet of type Y wood are needed?
(A) 2.4
(B) 3.6
(C) 6.0
(D) 7.2
(E) 8.4
- Q6.** A plumber is mixing 20 gallons of a 15% sulfuric acid solution with 30 gallons of a 25% sulfuric acid solution. What is the concentration of sulfuric acid in the final mixture?
(A) 18%
(B) 19%
(C) 20%
(D) 21%
(E) 22%
- Q7.** A chef is making a sauce that requires a 3 : 5 ratio of oil to vinegar. If the chef uses 9 cups of oil, how many cups of vinegar are needed?
(A) 12
(B) 15
(C) 18
(D) 21
(E) 24
- Q8.** A contractor needs to make a 100-gallon mixture of paint that is 40% red and 60% blue. If the contractor already has 20 gallons of red paint, how many gallons of blue paint are needed?
(A) 48
(B) 50
(C) 52
(D) 60
(E) 64

Open Ended Questions (FRQ)

- Q9.** A recipe for making cookies requires a 2 : 3 ratio of sugar to flour. If 4 cups of sugar are used, how many cups of flour are needed? Show your work and explain your answer.
- Q10.** A builder is making a 50-gallon mixture of cement that is 30% type A and 70% type B. If type A cement costs 1.80 per gallon and type B cement costs 2.40 per gallon, how much will the 50-gallon mixture cost? Show your work and explain your answer.